Embedded Control for Robotics

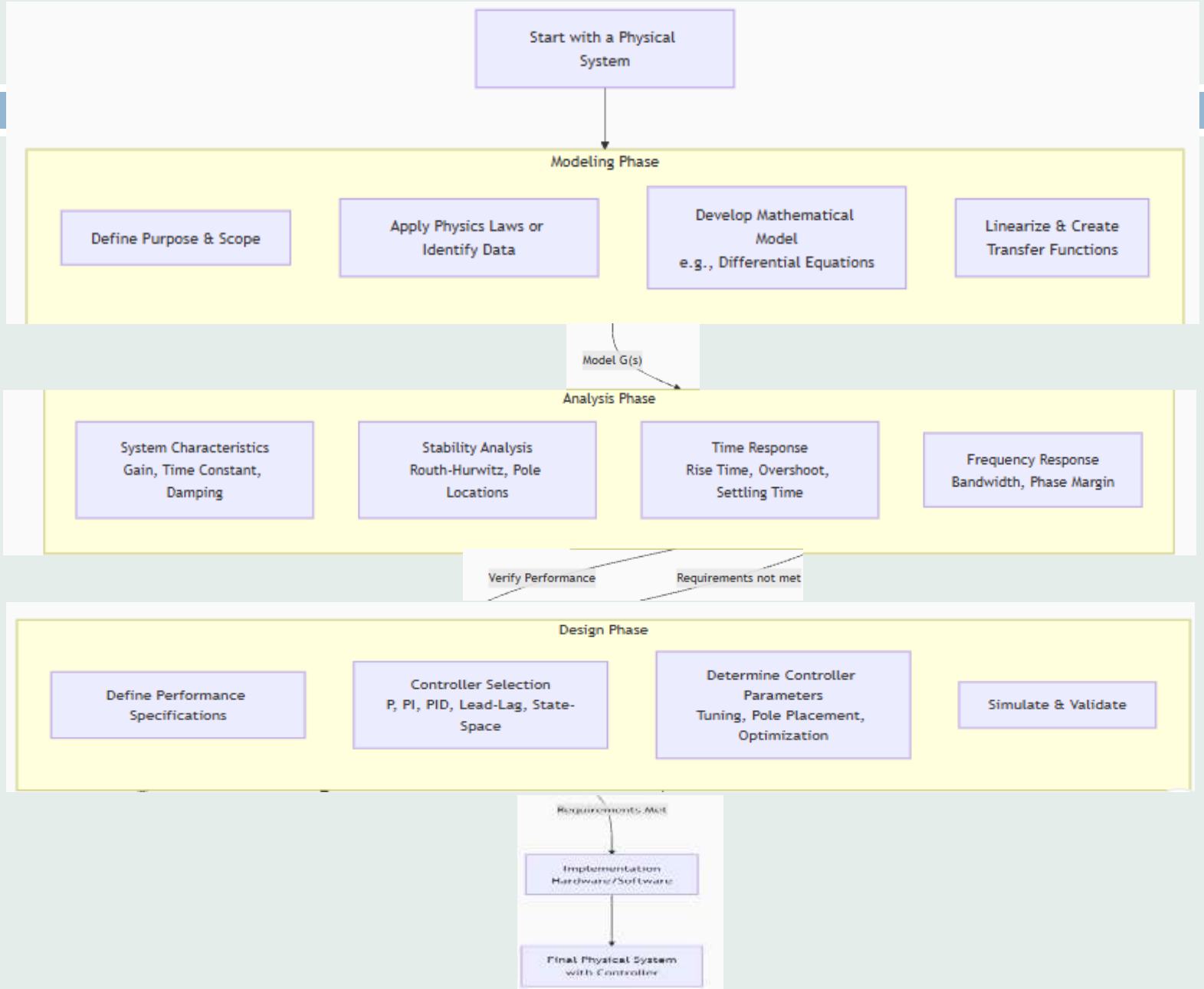
Lecture # 5c **Frequency Response & Bode Plot**

- Book:

Dr. Suleman Khan
CS, NUCES-FAST, Islamabad

Control Systems: Big Picture

1. Modeling
2. Analysis
3. Design



Frequency Analysis of Transfer Function

1. **Frequency Analysis**

2. **Input Signal**

$$u(t) = 1.0\sin(\omega t)$$

3. **Tool used is Bode plot**

- Why??
  - Phase Shift
  - Amplitudes

Frequency Response

For any transferfunction *G(s)* with the input

$$u(t) = 1.0\sin(\omega t)$$

will give the output

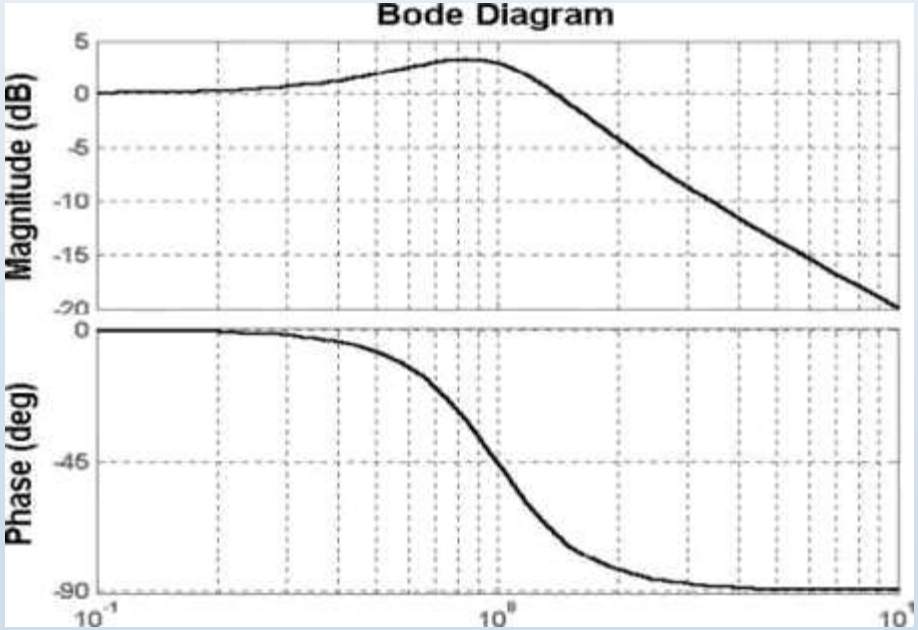
$$y(t) = a\sin(\omega t + \phi)$$

With the gain,

$$a = |G(j\omega)|$$

and the phase

$$\phi = \tan^{-1}\left(\frac{\text{imag}G(j\omega)}{\text{real}G(j\omega)}\right)$$



Magnitude decibels= 20 log| (G(juJ)|

Bode Plot?

The Bode plot is the frequency response plot of the transfer function of a system.

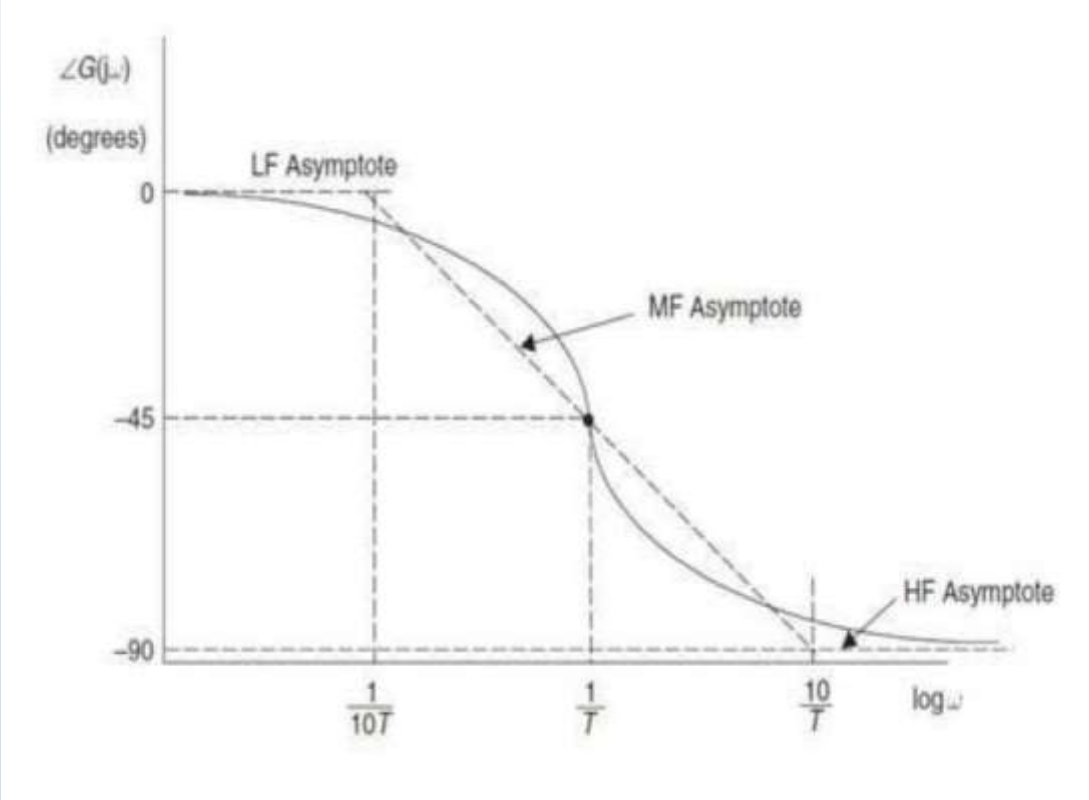
Bode plot consists of two graphs:

- ✓ One is the plot of magnitude of sinusoidal transfer function versus log ω.
- ✓ The other is a plot of the phase angle of a sinusoidal function versus log ω.

Phase vs frequency log

A section of TF can be represented as straight lines =asymptotic approximation

- A LF horizontal asymptote at 0
- A HF horizontal asymptote at -45
- A mid-frequency asymptote that intersects between the HF and LF asymptote -90



Frequency Response of First-order System

Two ways of writing:

$G(s) = \frac{k}{s+a}$ Good for frequency domain,

$G(s) = \frac{k}{\tau s+1}$ Good for time domain

Characteristics are:

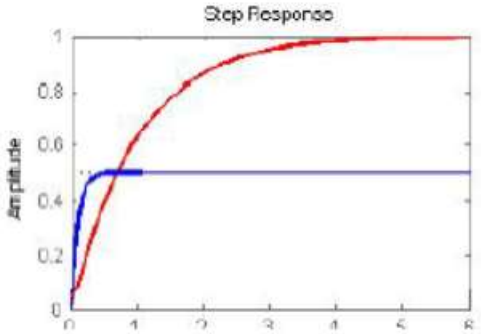
Pole

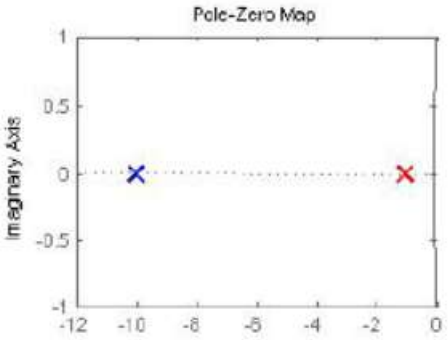
Dc-gain k

Time constant τ

Cut-off frequency

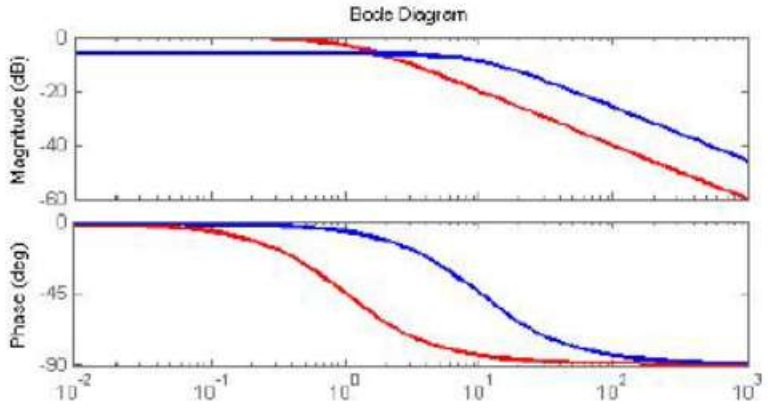
Phase lag at high freq.





Example: $G_1 = \frac{1}{s+1}$

$G_2 = \frac{5}{s+10}$



Second-order System

Model:

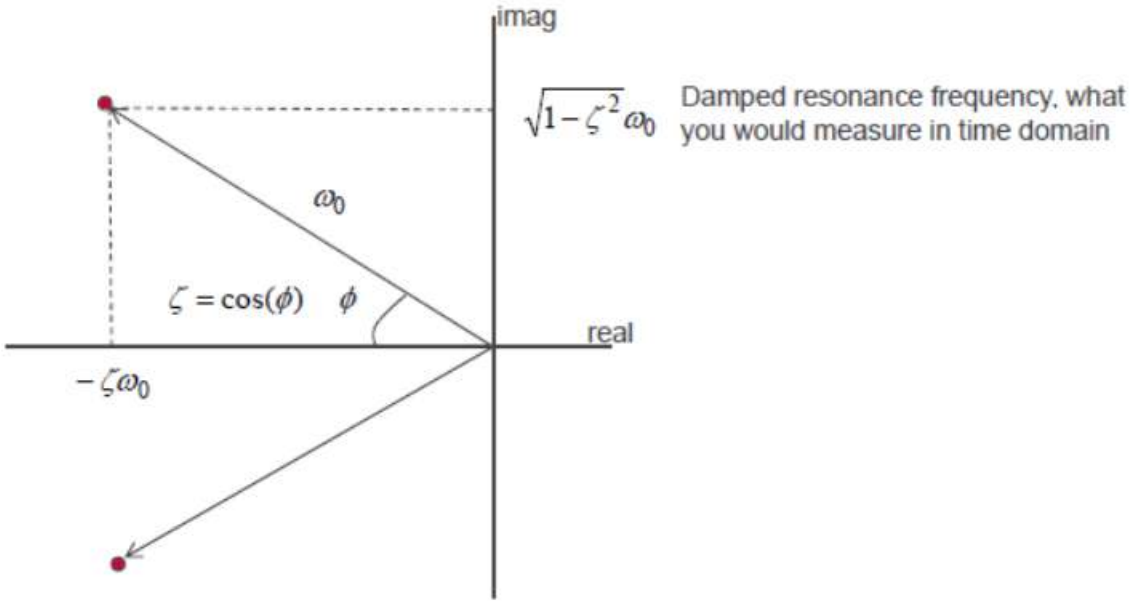
$$G(s) = \frac{\omega_0^2}{s^2 + 2\zeta\omega s + \omega_0^2}$$

Poles, complex conjugate when: $\zeta < 1$

$$s = -\zeta\omega_0 \pm \sqrt{\zeta^2 - 1}\omega_0 = -\zeta\omega_0 \pm j\sqrt{1-\zeta^2}\omega_0$$

Un-damped resonance frequency:

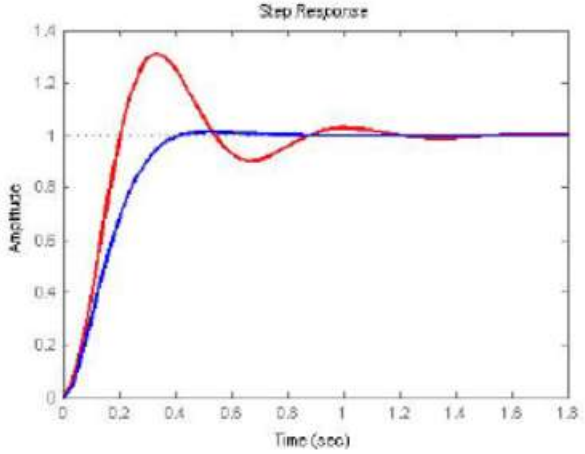
$$|s| = \sqrt{(\zeta\omega_0)^2 + (1-\zeta^2)\omega_0^2} = \omega_0$$

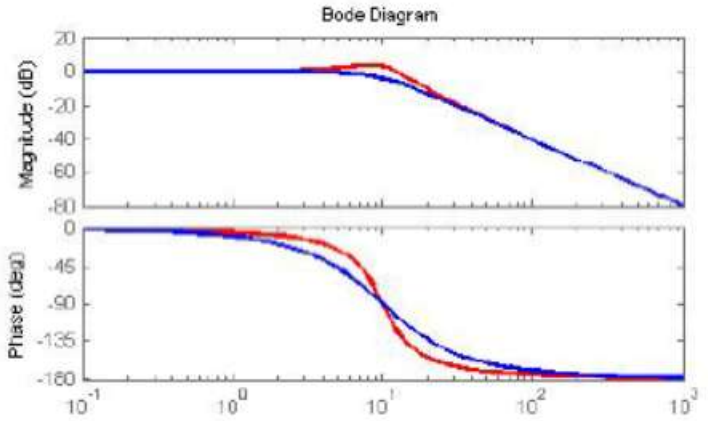


Damped resonance frequency, what you would measure in time domain

Step/Frequency Response

Two models with same frequency but different damping



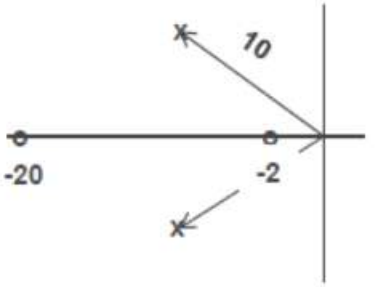


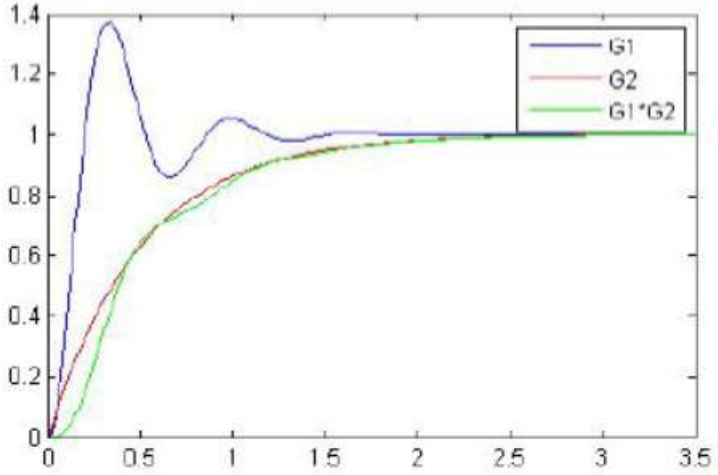
Low pass characteristics, 180 degree phase shift at high frequencies
Overshot, What is the damping ratio ?

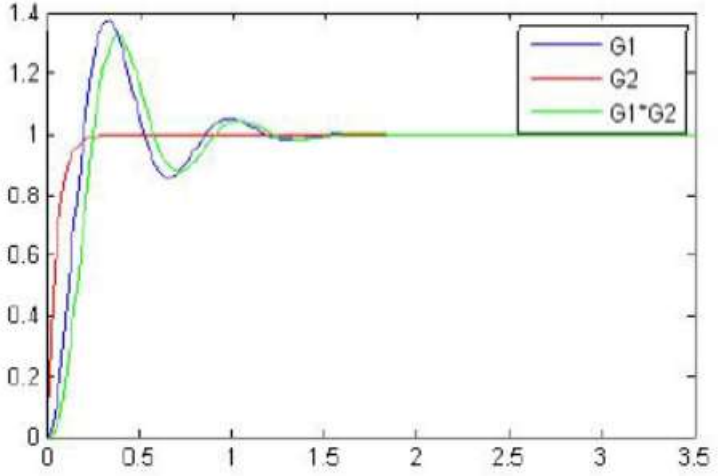
A second order model $G_1(s) = \frac{100}{s^2 + 6s + 100}$ is superpositioned

with a first order model $G_2(s) = \frac{a}{s + a}$, such that $G_s(s) = G_1(s)G_2(s)$

In left figure is a=2 and in right figure a=20



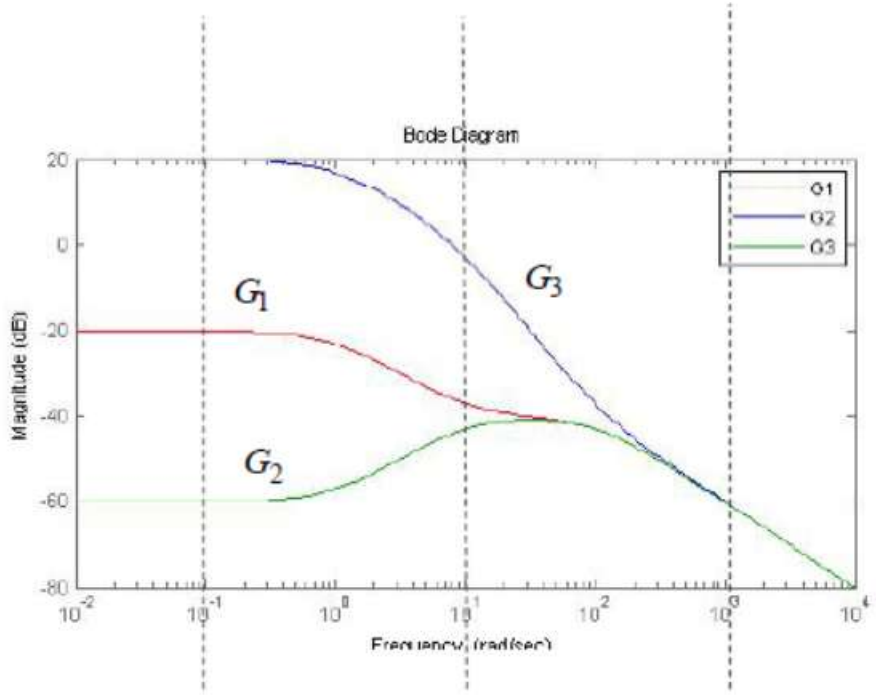




The TF with the slowest pole dominates the step response

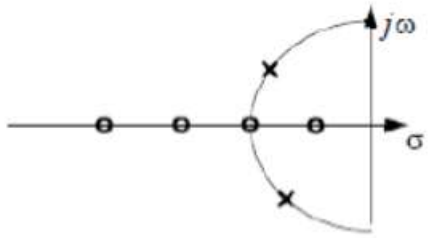
Pole impact on frequency

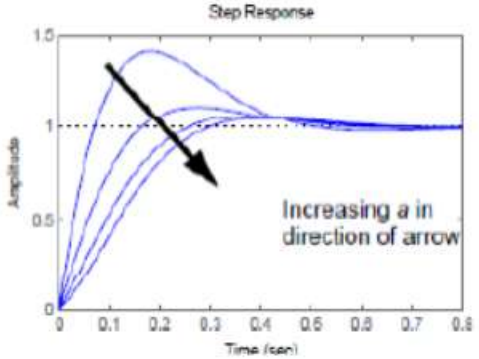
$$G_1(s)=\frac{s+10}{(s+1)(s+100)},\quad G_2(s)=\frac{s+100}{(s+1)(s+10)},\quad G_3(s)=\frac{s+1}{(s+10)(s+100)}$$

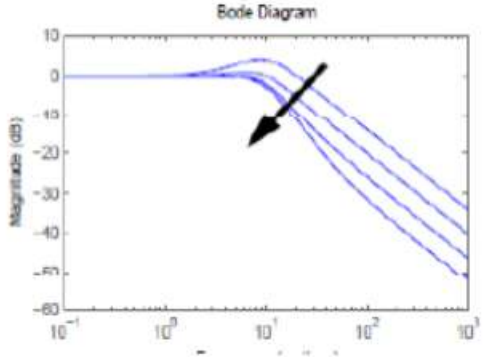


Zero impact on frequecy

$$G(s) = \frac{\left(\frac{s}{\omega_0 a} + 1\right)\omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2} \quad a = [0.5, 1, 2, 4]$$



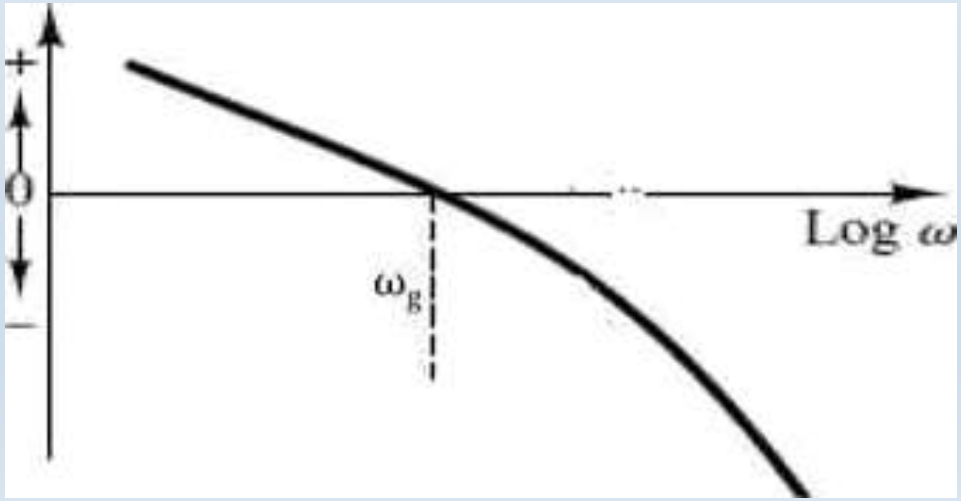




Gain Crossover frequency

The gain crossover frequency

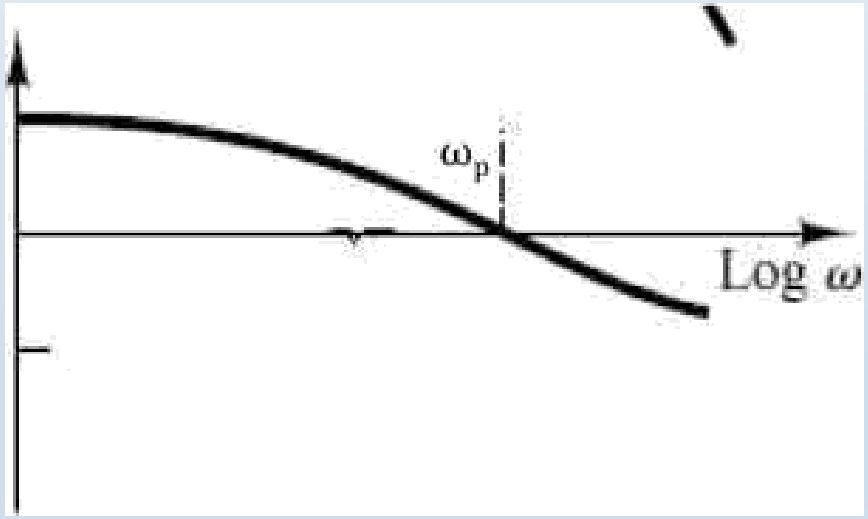
- The frequency at which the magnitude of the open loop transfer function, |G(j‹c)| is unity.
- In logarithmic scale this corresponds to zero db



Phase Cross over frequency

The phase crossover frequency

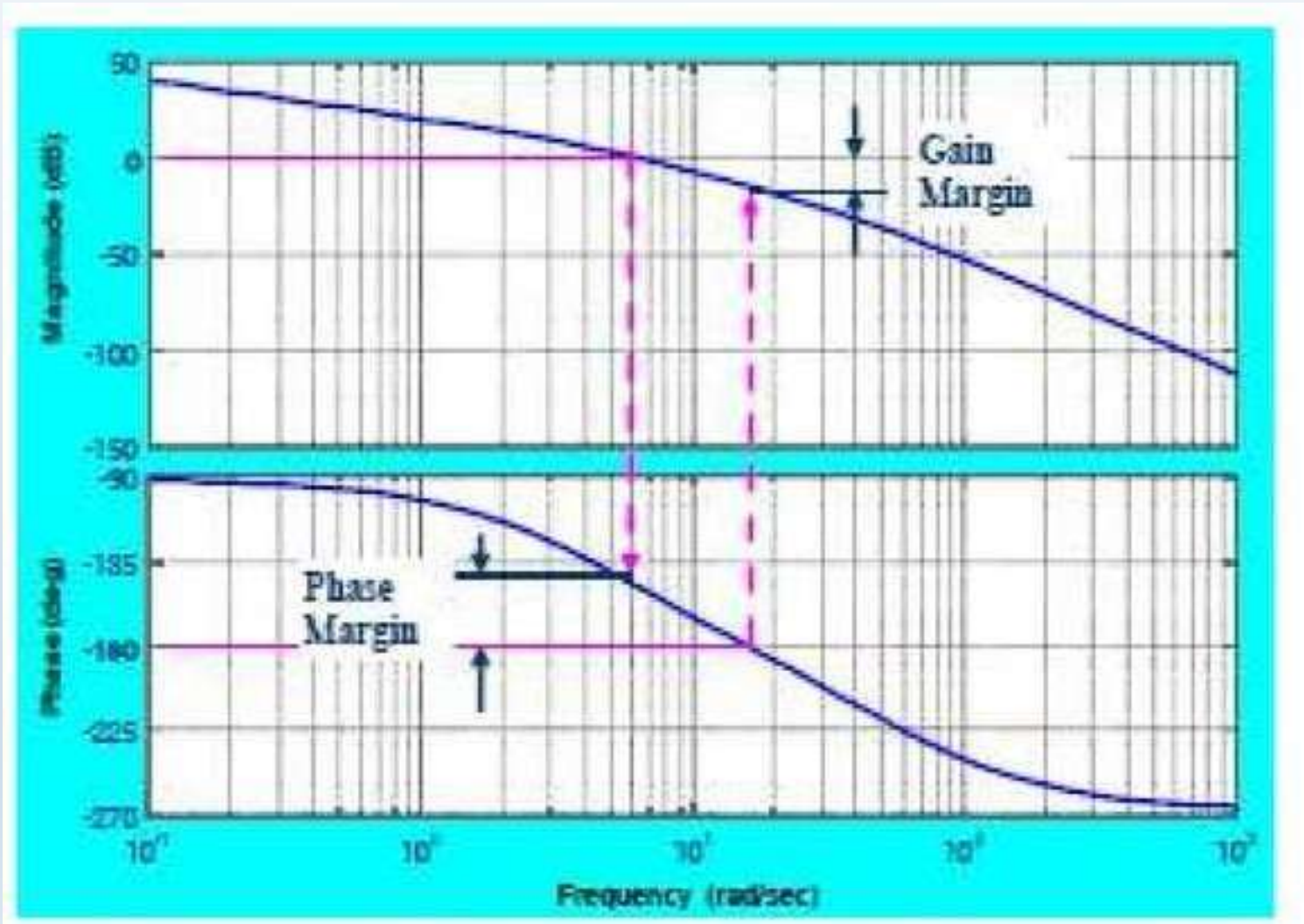
- the frequency at which the phase of the open loop transfer function, zG(jw) is -180°.



Stability Analysis

- If the phase cross over frequency ω_{pc} is greater than the gain cross over frequency ω_{gc}, then the control system is **stable**.
- If the phase cross over frequency ω_{pc} is equal to the gain cross over frequency ω_{gc}, then the control system is **marginally stable**.
- If the phase cross over frequency ω_{pc} is less than the gain cross over frequency ω_{gc}, then the control system is **unstable**.

Phase Margin and gain Margin



Gain margin

- The gain margin is the amount of additional gain required at the phase crossover frequency to bring the system to the verge of instability.
- The gain margin is the reciprocal of the magnitude $|G(j\omega)|$ at the phase crossover frequency.

$$GM = \frac{1}{|G(j\omega_p)|}$$

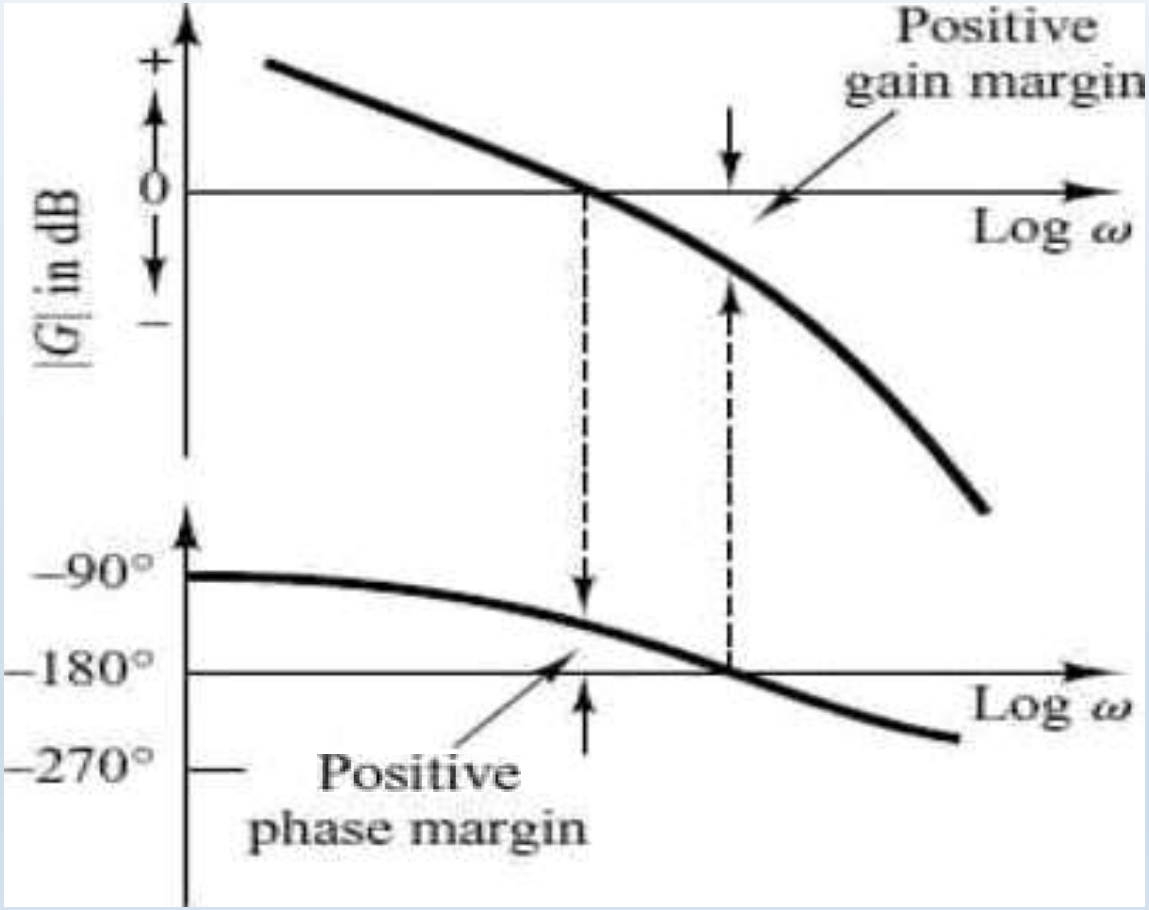
In decibels, $GM = -20\log |G(j\omega_p)|$

Phase margin

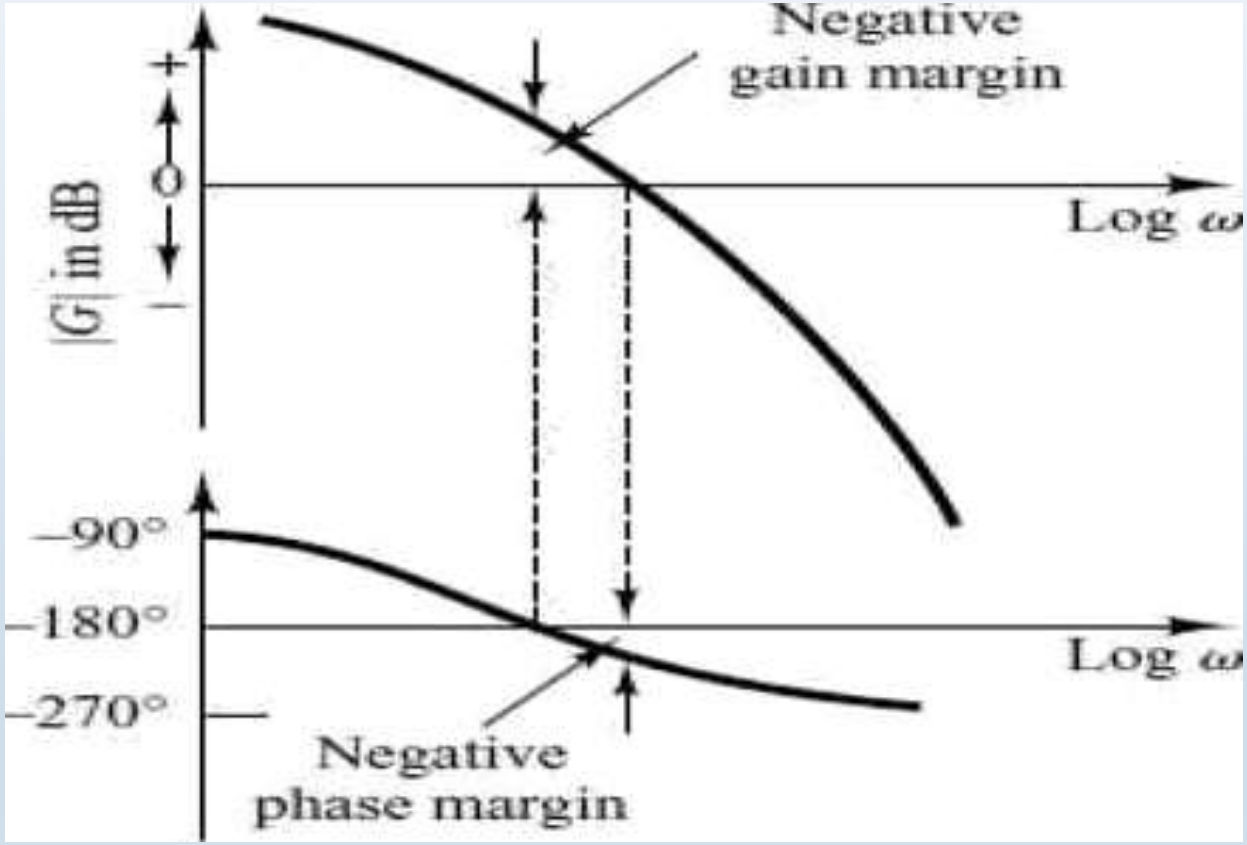
- The phase margin is the amount of additional phase lag required at the gain crossover frequency to bring the system to the verge of instability.
- The phase margin is 180° plus the phase angle of the open-loop transfer function at the gain crossover frequency ω_g

$$PM=180°+Ø$$

Stable System



Unstable System



Stability Analysis

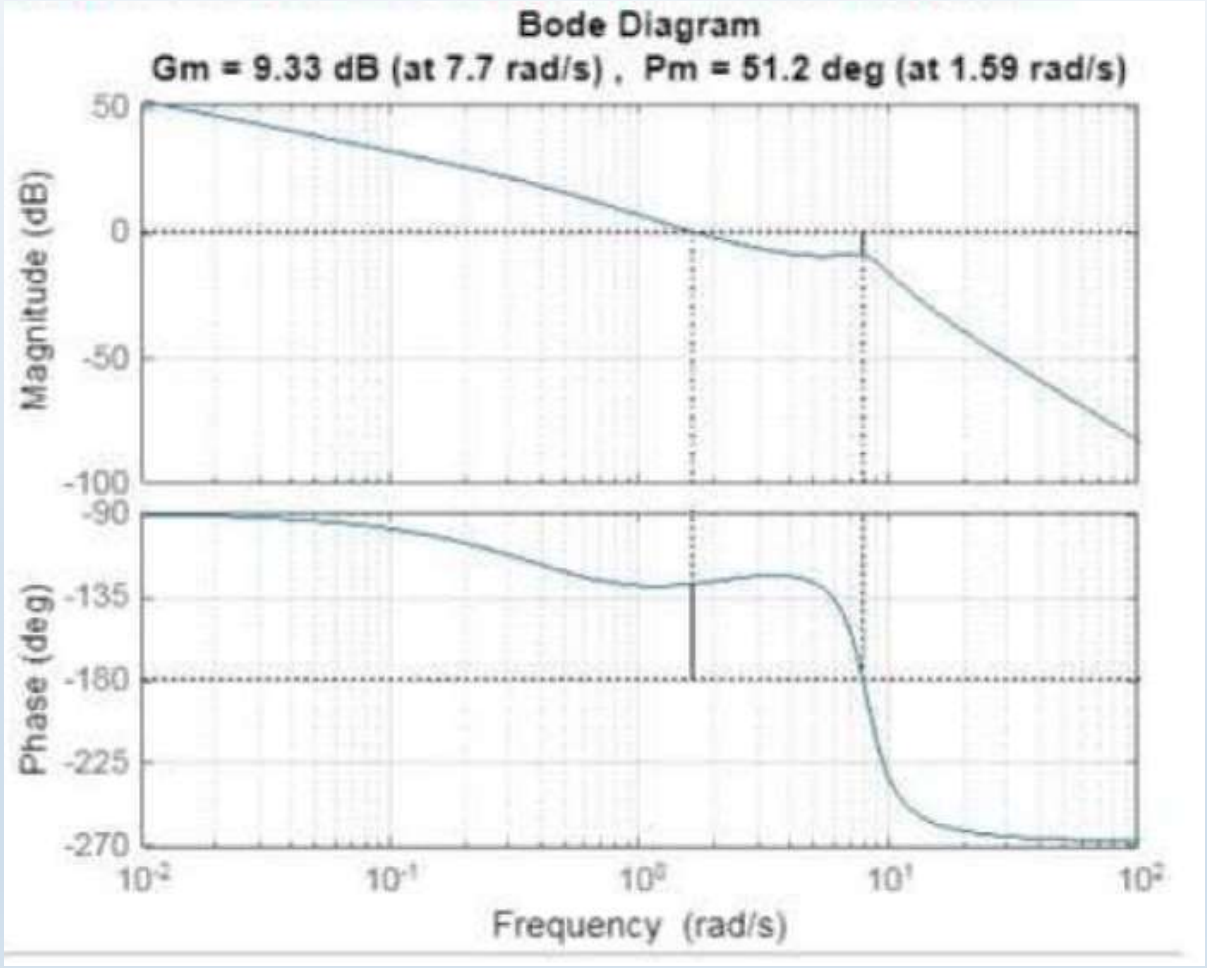
1. If both the gain margin GM and the phase margin PM are positive, then the control system is stable.
2. If both the gain margin GM and the phase margin PM are equal to zero, then the control system is marginally stable.
3. If the gain margin GM and/or the phase margin PM are/is negative, then the control system is **unstable.**

Exp # 1

$$\frac{64(s+2)}{s(s+0.5)(s^2+3.2s+64)}$$

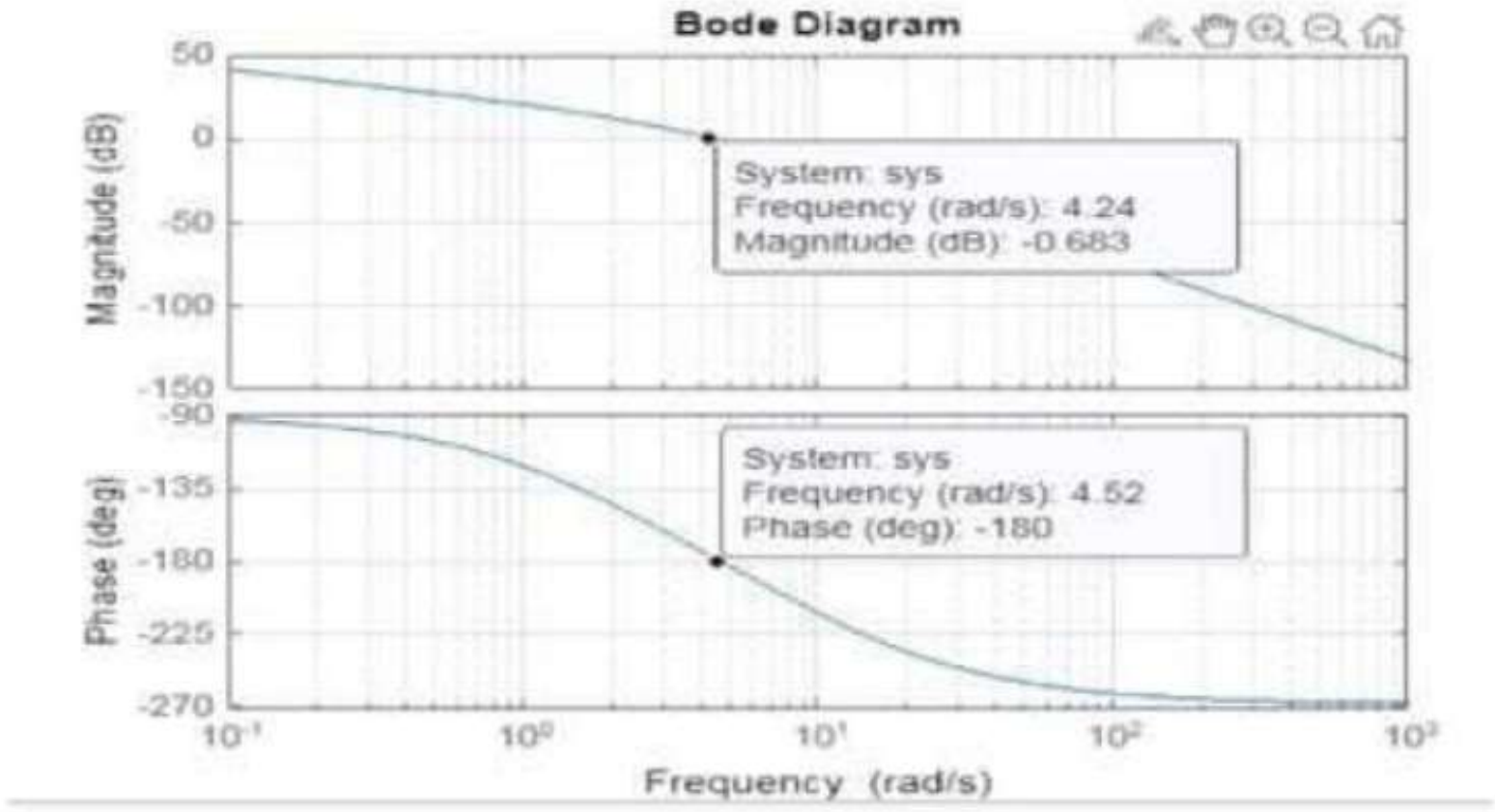
```
num=[64 128];
den=[1 3.7 65.6 32 0]
bode(num,den)
margin(num,den)
```

Stable?



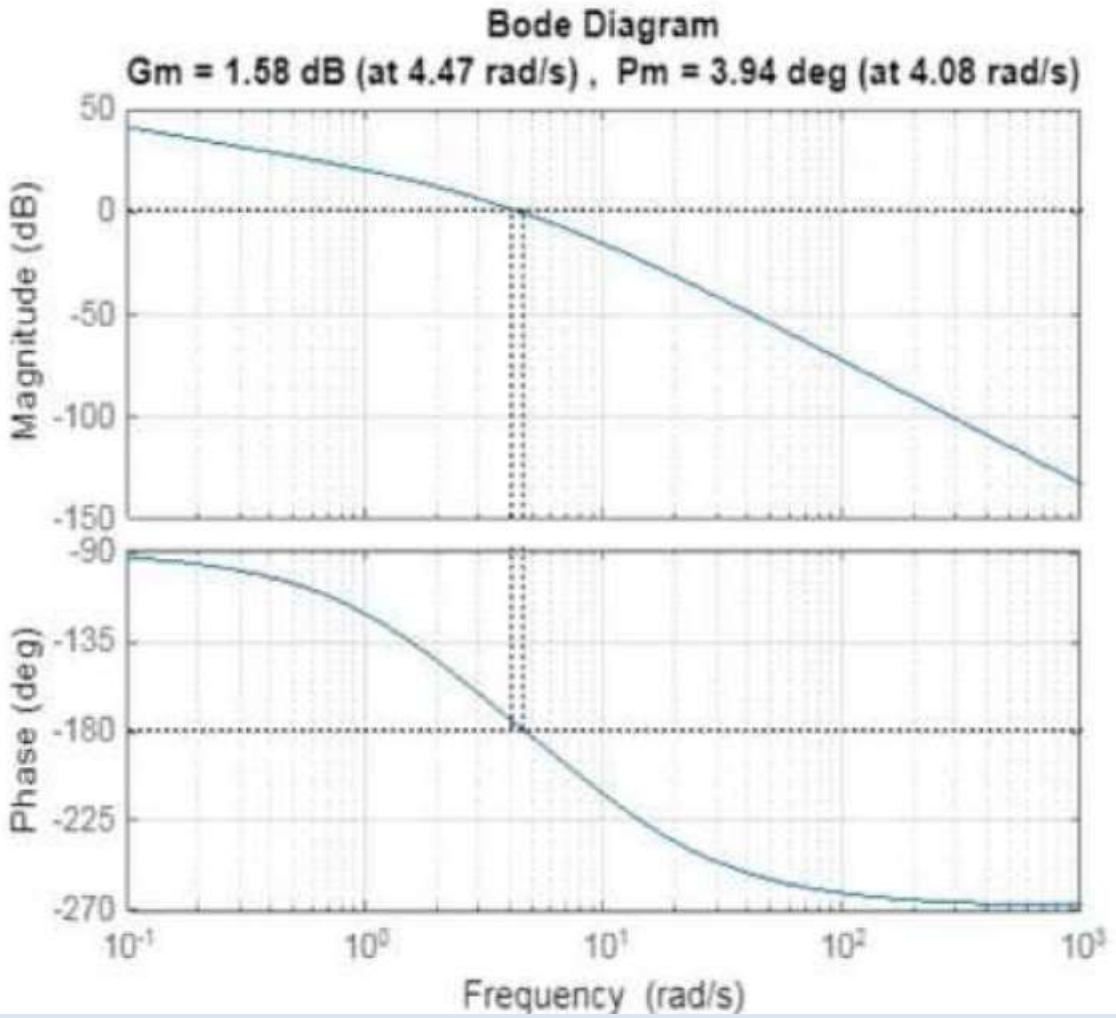
Exp # 2

$$\frac{10}{s(1+0.5s)(1+0.1s)}$$



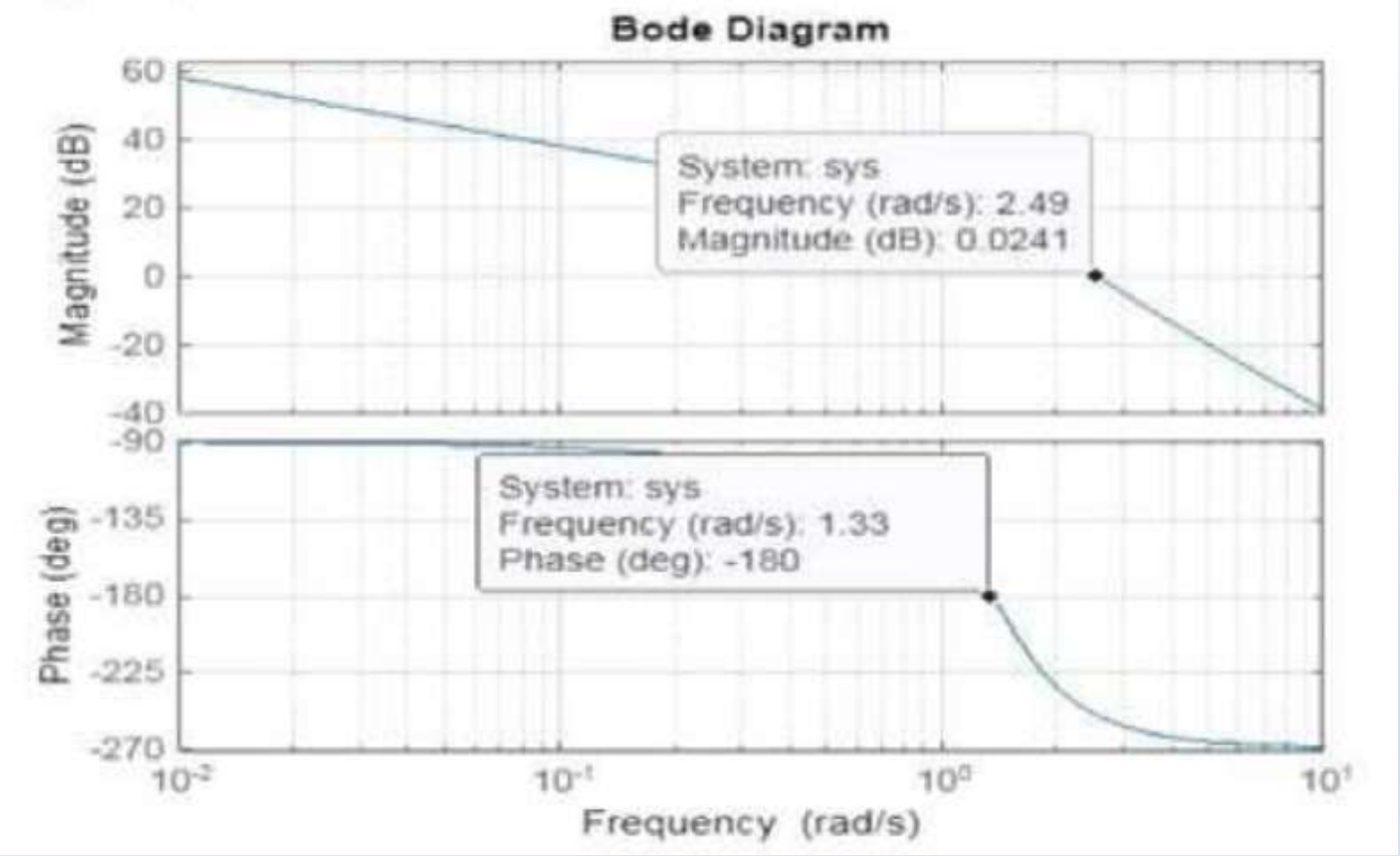
Continue

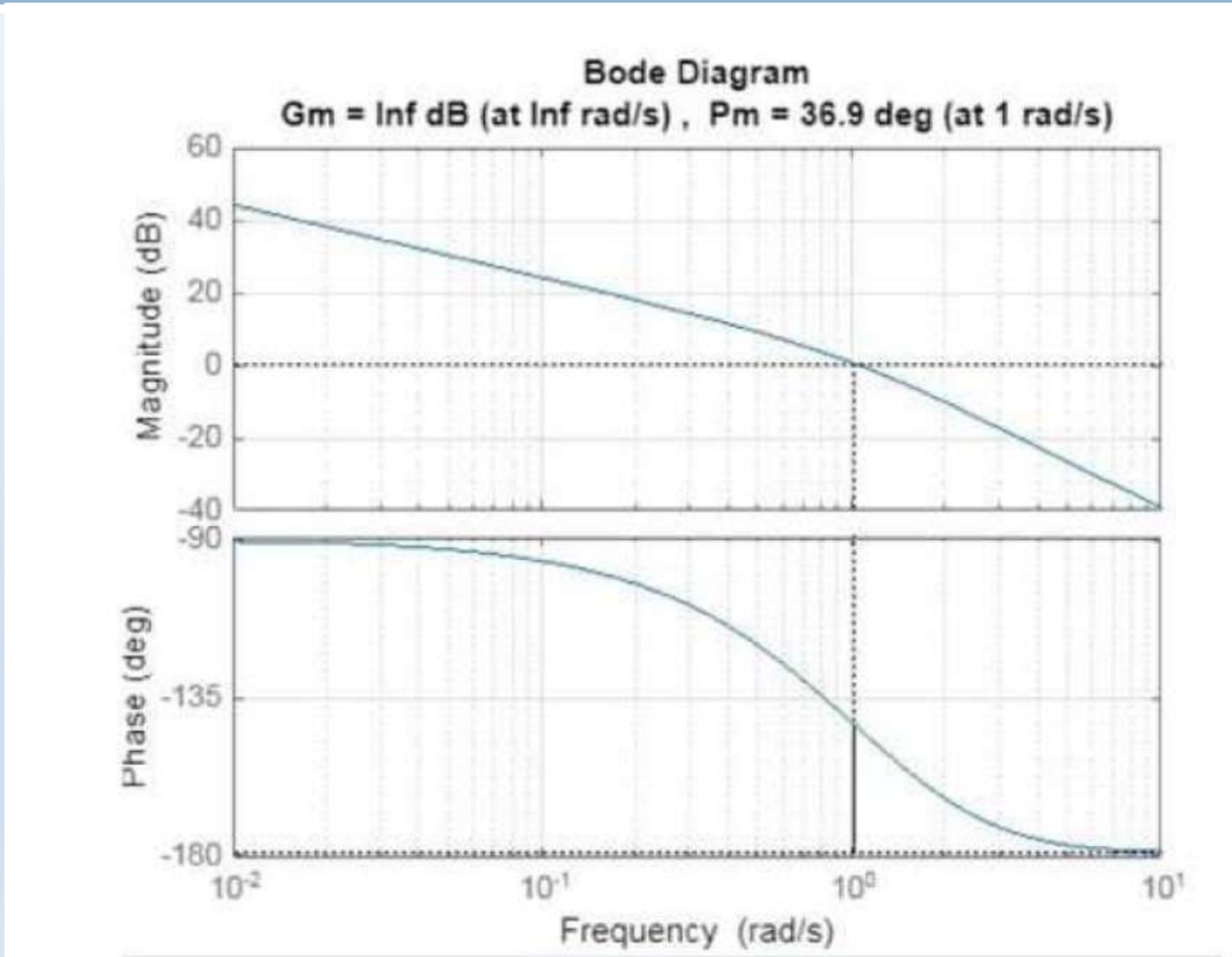

```
num=[10];
den=[0.05 0.6 1 0];
bode(num,den)
margin(num,den)
grid on
```



Exp # 3

$$\frac{(s+3)}{s(s+1)(s+2)}$$




Bode Diagram
Gm = Inf dB (at Inf rad/s) , Pm = 36.9 deg (at 1 rad/s)
Magnitude (dB)
60
40
20
0
-20
-40
Phase (deg)
-90
-135
-180
10^{-2}
10^{-1}
10^{0}
10^{1}
Frequency (rad/s)

Poles and Zero Relation with bode

Type of term	G(jω)H(jω)	Slope(dB/dec)	Magnitude (dB)	Phase angle(degrees)
Constant	K	0	20logK	0
Zero at origin	jω	20	20logω	90
'n' zeros at origin	$(j\omega)^n$	20n	20nlogω	90n
Pole at origin	1/jω	−20	−20logω	−90or270
'n' poles at origin	$(1/j\omega)^n$	−20n	−20nlogω	−90n or 270n
Simple zero	$1+j\omega_r$	20	0 for $\omega < 1/r$ $20\log\omega_r$ for $\omega > 1/r$	0 for $\omega < 1/r$ 90 for $\omega > 1/r$
Simple pole	$1/(1+j\omega_r)$	-20	0 for $\omega < 1/r$ $-20\log\omega_r$ for $\omega > 1/r$	0 for $\omega < 1/r$ -90 or 270 for $\omega > 1/r$

Continue…

Type of term	G(jω)H(jω)	Slope (dB/dec)	Magnitude (dB)	Phase angle(degrees)
Second order derivative term	$\omega_n^2 (1-(\omega^2/\omega_n^2) + (2j\delta\omega/\omega_n))$	40	$40\log\omega_n$ for $\omega < \omega_n$ $20\log(2\delta\omega_n^2)$ for $\omega = \omega_n$ $40\log\omega_n$ for $\omega > \omega_n$	0 for $\omega < \omega_n$ 90 for $\omega = \omega_n$ 180 for $\omega > \omega_n$
Second order integral term	$\dfrac{1}{\omega_n^2 (1-(\omega^2/\omega_n^2) + (2j\delta\omega/\omega_n))}$	-40	$-40\log\omega_n$ for $\omega < \omega_n$ $-20\log(2\delta\omega_n^2)$ for $\omega = \omega_n$ $-40\log\omega_n$ for $\omega > \omega_n$	0 for $\omega < \omega_n$ -90 for $\omega = \omega_n$ -180 for $\omega > \omega_n$

Calculate mag and phase

- The **magnitude** of the open loop transfer function in dB is -

$$M=20\log|G(j\omega)H(j\omega)|$$

- The **phase angle** of the open loop transfer function in degrees is -

$$\phi=\angle G(j\omega)H(j\omega)$$

Note – The base of logarithm is 10.

Assignment # 3 Python code
Due date

Determine the transfer function of the R-C network

- **Transfer function.**
- **Plot the Pole-zero plot**
- **Step Response**
  - **DC gain/Time constant**
- **Bode plot**
  - **Phase margin**
  - **Gain margin**

